

Hospital Patient Records Analytics

End-to-end data cleaning, analysis, and visualization
of Massachusetts General Hospital patient data | 2011–2022

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974

Patients

27,891

Encounters

47,701

Procedures

DATASETS — 5 FILES

Patients 974 rows

Demographics, age, race, gender, location

Encounters 27,891 rows

Visit records by type, cost, payer, duration

Procedures 47,701 rows

Procedures, base cost, duration, reason codes

Payers 10 rows

10 insurance providers including Medicare

Organizations 1 row

Massachusetts General Hospital, Boston MA

DATA CLEANING

- Removed numeric artifacts from patient names using regex
- Fixed ZIP codes: 4-digit floats → zero-padded 5-digit strings
- Parsed datetime fields; computed duration (mins)
- Calculated patient age from birth / death dates
- Created full name column (first + maiden + last)
- Filled NaN reason codes with 0; descriptions with 'Unknown'
- Dropped SUFFIX column (only 21 / 974 populated)
- Exported cleaned CSVs for Power BI ingestion

4 ANALYSIS QUESTIONS

Q1 How many patients were admitted or readmitted over time?

Q2 How long do patients stay on average, by encounter type?

Q3 What is the average cost per visit, by encounter type?

Q4 How many procedures are covered by insurance?

974

Unique patients

Boston MA, 2011–2022

70.99

Average patient age

Years at time of record

32.97

Avg procedure
duration

Minutes per procedure

60.8%

Elderly patients

Age 65+ (592 of 974)

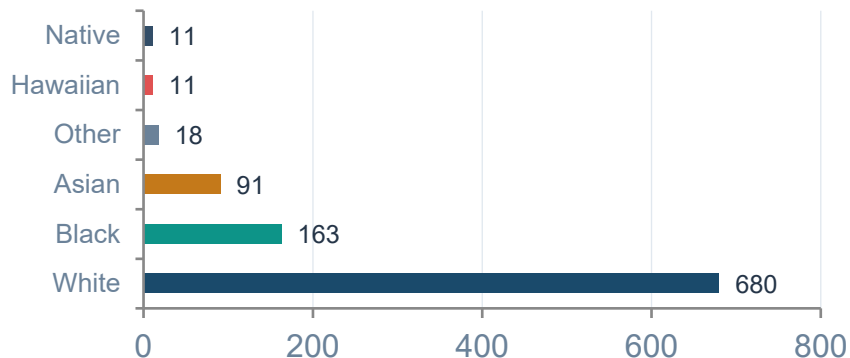
7h 15m

Avg hospital stay

Across all encounter types

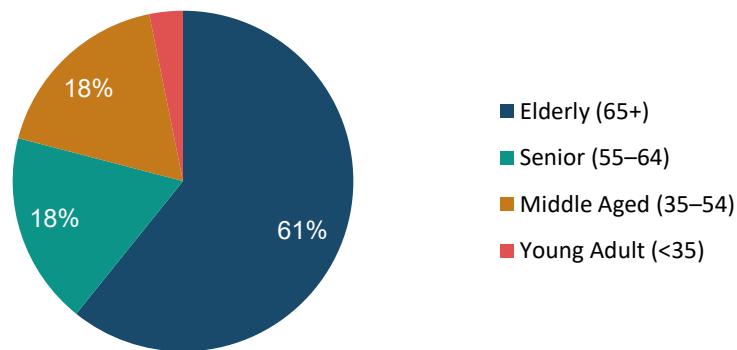
Patient count by race

680 white · 163 black · 91 Asian · smaller groups



Age distribution of patients

Classified into 4 age cohorts from notebook analysis



27,891

Total encounters

Jan 2011 – Jan 2022

17,366

Readmissions

Within 30-day window

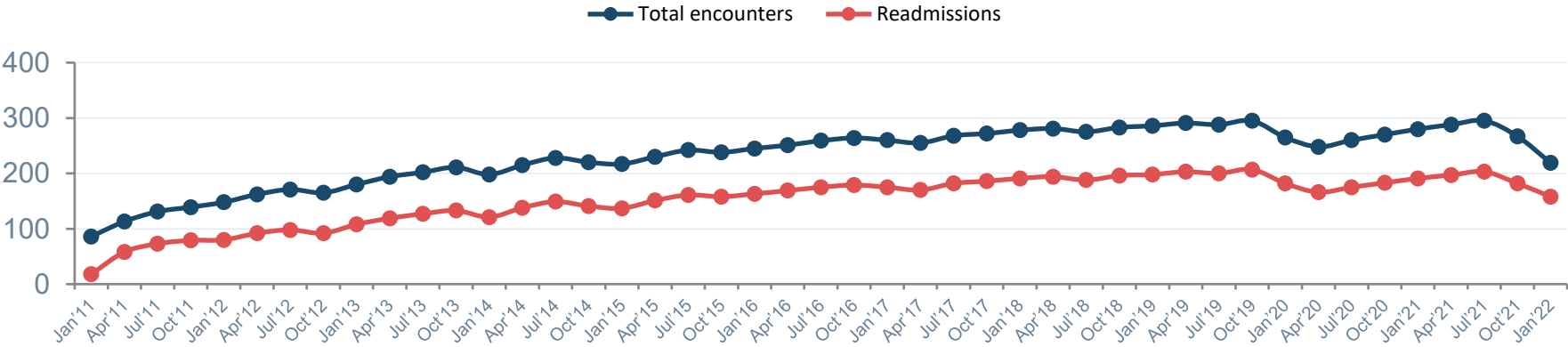
987

First admissions

Unique first visits

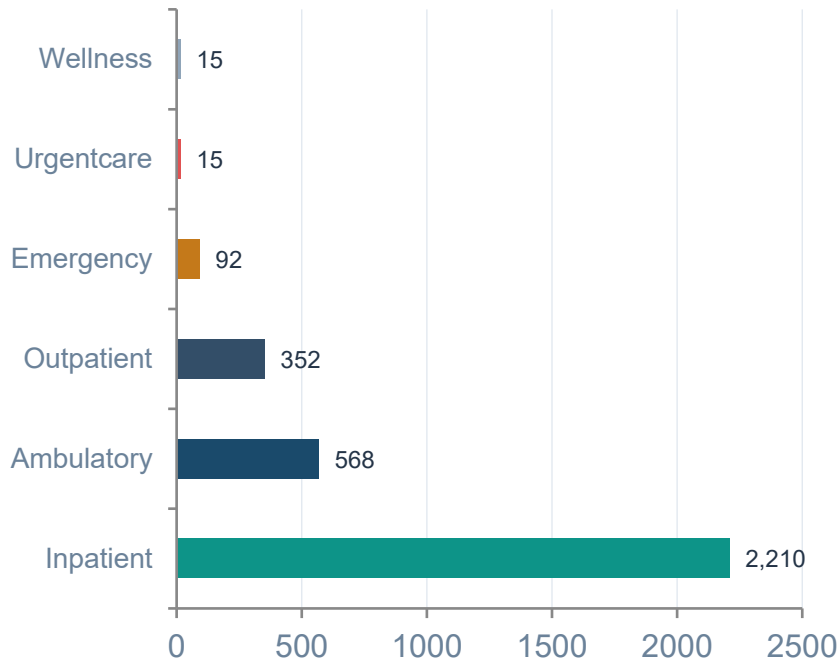
Monthly encounters vs. readmissions — 2011–2022

Readmission = same patient returning within 30 days · Total readmissions: 17,366



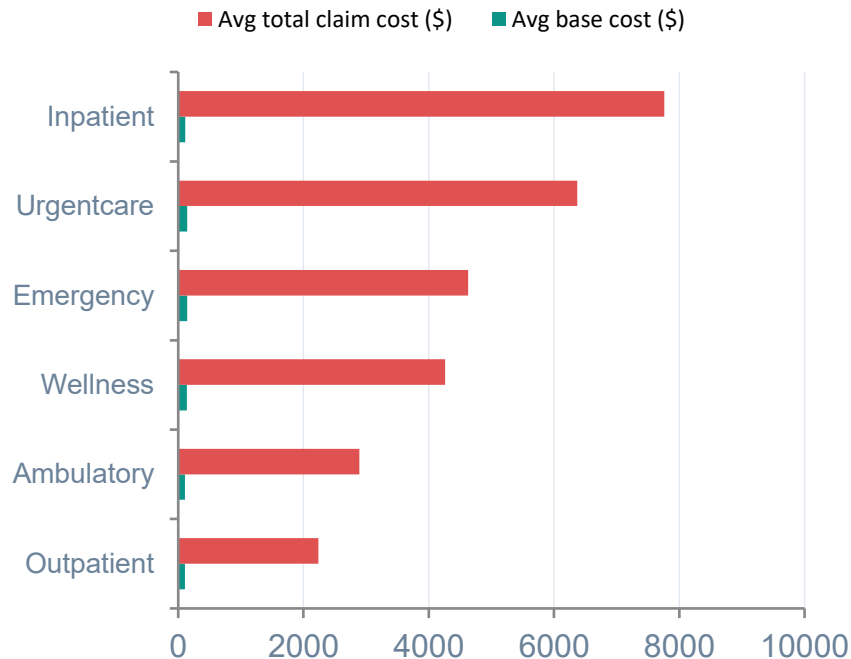
Avg length of stay by encounter type

Overall average: 7h 15min · Inpatient is 2.4× longer than outpatient



Avg encounter cost vs. total claim cost

Base cost avg \$116 · Total claim avg \$3,640 — inpatient most expensive



138

Procedures covered

Unique procedure codes w/ payer coverage

\$1,115

Avg payer coverage

Per encounter across all payers

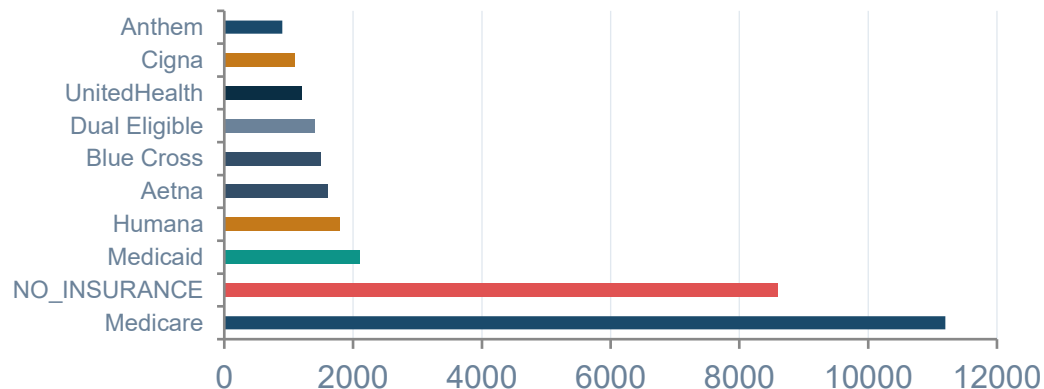
No insurance

2nd largest group

Only Medicare sees more procedures

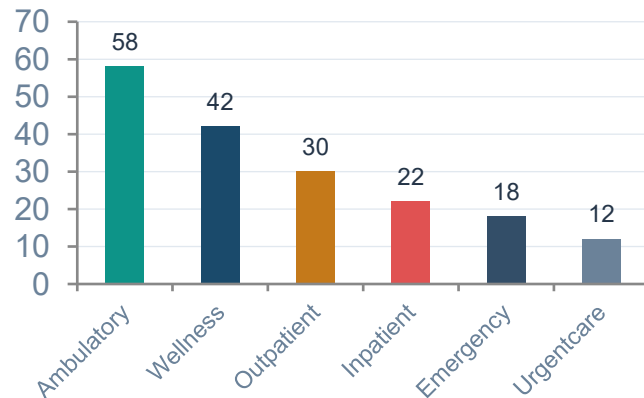
Procedure count by insurance provider

Medicare dominant — NO_INSURANCE second-largest group signals coverage gap



Covered procedures by encounter type

Ambulatory carries the most coverage volume



Power BI Dashboard

01

Patient demographics

Race breakdown, gender split, geographic concentration in Suffolk County, Boston.

02

Procedure & payer volume

Medicare leads with 11K+ procedures. NO_INSURANCE second — a population health signal.

03

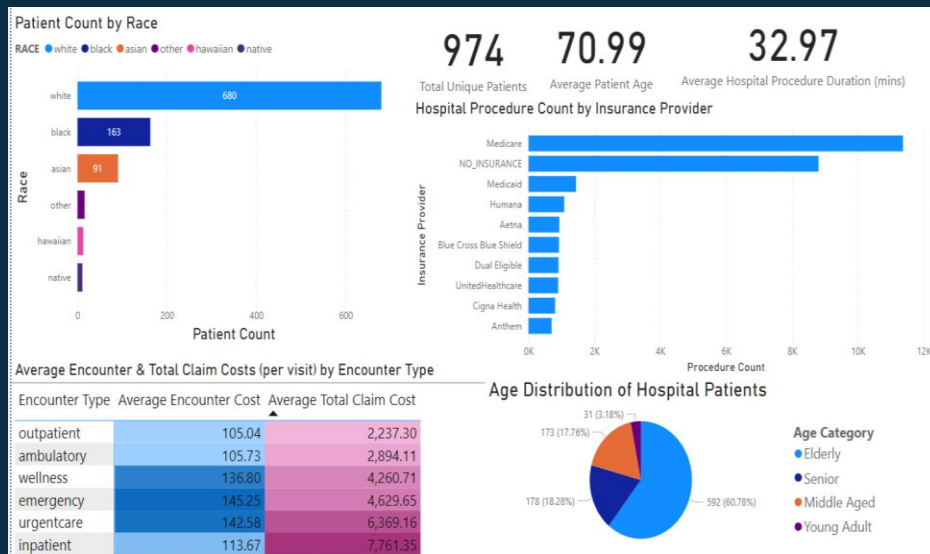
Cost by encounter type

Inpatient highest at avg \$7,761 total claim; urgentcare and wellness at opposite ends.

04

Age cohort distribution

60.8% elderly patients (592 of 974), reinforcing the Medicare dominance in payer mix.



Four data-driven actions grounded in the findings — mapped to the stakeholders who own them.

01

Hospital Operations

Launch a chronic disease readmission program

62% of all encounters are return visits within 30 days. Prioritize care coordination for high-frequency readmission patients — particularly those with chronic conditions — to reduce preventable returns and free up bed capacity.

02

Revenue Cycle & Finance

Screen uninsured patients for coverage eligibility

Uninsured patients are the second-largest procedure group. Given the predominantly elderly patient population, many likely qualify for Medicare or Medicaid. A point-of-entry eligibility check could improve both patient access and revenue recovery.

03

Clinical Leadership

Review inpatient care pathways for efficiency

Inpatient stays average 36h 50min at \$7,761 per claim — significantly above other encounter types. Audit top inpatient diagnoses against clinical benchmarks to identify where length-of-stay can be safely reduced without compromising outcomes.

04

Executive Leadership

Invest in ongoing population health monitoring

This analysis captures a snapshot. The trends here — rising encounter volume, aging patient mix, heavy Medicare dependence — require quarterly tracking. Commission a live dashboard to surface early warning signals before they become operational crises.